1 PROBLEM IDENTIFICATION AND MOTIVATION

- Identify accessibility barriers in blockchain healthcare interfaces for people with disabilities (PwDs).
- Review blockchain healthcare, HCI, accessibility, and assistive technology literature.
- Define the research gap, objectives, and research questions (RQ1–RQ3).

2 OBJECTIVES OF A SOLUTION (DESIGN)

- Derive accessibility requirements and HCI design principles (AAD, PTD, CID, and UAC).
- Develop the BCHAIN-ACCESS framework with four integrated perspectives.
- Define the system architecture and design specifications.

3 DESIGN AND DEVELOPMENT

- Develop the BCHAIN-ACCESS reference prototype.
- Implement the four-layer system architecture.
- Integrate accessibility features and inclusive user interfaces.

4 DEMONSTRATION AND EVALUATION

- Perform automated accessibility evaluation using Lighthouse, WAVE, and Axe DevTools.
- Conduct expert evaluation using heuristic evaluation, cognitive walkthrough, SUS, and an accessibility questionnaire (N = 10).
- Analyze results and validate the proposed framework.

5 EVALUATION AND COMMUNICATION

- Validate the framework's usability, accessibility, and technical feasibility.
- Discuss findings, limitations, and future research directions.
- Disseminate the framework, reference prototype, and design guidelines.

OUTPUTS

- **BCHAIN-ACCESS Framework**
- **Reference (Prototype)**
- **Expert Validation**
- **Accessibility Design Guidelines**

Iterative Feedback Loop

(Refinement based on evaluation results and expert feedback)

KNOWLEDGE BASE

- **Blockchain Healthcare Literature**
- **HCI, UI/UX and Accessibility Literature**
- **Accessibility & Assistive Tech Literature**
- **Standards and Guidelines (WCAG, GDPR, EN 301)**
- **Tools and Platforms (Lighthouse, WAVE, Fabric)**